Case B

Mr. O'Shea — 72-Year-Old Retired Librarian | Working Diagnosis: Osteoporotic Vertebral Compression Fracture (T7–T8)

Target: 70%+ | Musculoskeletal (MSK) Physiotherapy

■ PROBLEM LIST

- Mid-thoracic pain VAS 4/10 — worse with prolonged standing/loading
- Progressive thoracic kyphosis (vertebral body wedging)
- Tenderness and percussion pain at T7–T8 (fracture site)
- Height loss ("shrinking") — loss of vertebral body height
- Likely reduced thoracic and hip extension ROM
- Significant secondary osteoporosis (corticosteroid-induced)
- Likely reduced back extensor and core muscle strength
- Inability to stand for prolonged periods (shopping, standing at kitchen)
- Difficulty with activities requiring thoracic loading (lifting, carrying)
- Possible difficulty with dressing (bending forward)
- Reduced walking tolerance and exercise capacity
- Potential social isolation secondary to pain and reduced mobility

■ RED FLAGS

- Bilateral lower limb weakness, paraesthesia, gait disturbance, or bladder/bowel dysfunction
- Severe, constant, unremitting back pain — worse lying down, disturbing sleep
- Fever, rigors, unexplained weight loss (>5 kg in 3 months), night sweats
- Sudden escalation in pain severity or new neurological symptoms following assessment
- Absent or decreasing reflexes in lower limbs + upper motor neuron signs (clonus, Babinski)

■ PATHOPHYSIOLOGY

- Osteoporosis is characterised by reduced bone mineral density (BMD) and disrupted trabecular microarchitecture, resulting in decreased bone strength and increased fracture susceptibility.
- The thoracic vertebrae (T7–T8) are vulnerable sites for osteoporotic VCF because the natural thoracic kyphosis directs axial compressive load predominantly onto the anterior vertebral body — the region with the least cortical bone support.
- Localised periosteal nerve endings and the sinuvertebral nerve at the fracture site generate deep, aching nociceptive pain, which is exacerbated by axial loading (standing, extension) and relieved by offloading (lying down).

■ DIFFERENTIAL DIAGNOSES

1. 1. Spinal Metastases / Pathological Fracture (Malignancy)

- 72-year-old male — age of peak malignancy incidence
- Mid-thoracic pain is a common site for metastatic deposits (lung, prostate, renal)
- Vertebral body destruction leading to pathological fracture can mimic VCF

2. 2. Scheuermann's Disease / Thoracic Disc Pathology

- Scheuermann's kyphosis presents with progressive thoracic kyphosis and mid-back pain
- Typically younger onset but may be longstanding, undiagnosed, and symptomatic in old age
- Disc degeneration contributes to thoracic pain and kyphosis in older adults

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Spinous Process Percussion Test

What it Measures: Bony tenderness and structural integrity of the vertebral bodies — used to localise spinal fracture or osseous pathology. | How to Perform: 1. Patient seated or prone. 2. Using a closed fist or a reflex hammer handle, apply direct percussive tap to each thoracic

Thoracic Kyphosis Measurement — Kyphosis Index / Flexicurve Ruler

What it Measures: Degree of thoracic kyphosis — quantifies the postural deformity caused by vertebral wedging and establishes a baseline for monitoring progression or improvement | How to Perform: 1. Patient standing in natural posture. 2. Identify and mark C7 and L1 spinous proc

Lower Limb Neurological Screen (Reflexes, Myotomes, Dermatomes) + Babinski Sign

What it Measures: Thoracic spinal cord integrity — screens for cord compression (myelopathy) from posterior VCF wall retropulsion or thoracic canal stenosis. | How to Perform: 1. Lower limb reflexes: Patellar (L3–L4), Achilles (S1–S2) — compare bilaterally with reflex hammer. 2.

■ TREATMENT PLAN

■ Initial Phase — Weeks 1–4 (Pain Management, Safety, Stabilisation)

- Pain understanding and pacing: "Your pain is coming from the weakened vertebrae in your mid-back. The good news is that with the right exercises and activity management, we can reduce your pain, strengthen your back muscles, and prevent further fractures. Movement is important — too much rest will make your bones and muscles weaker."
- Fall prevention — the most critical long-term message: "Because of your osteoporosis, preventing a fall is our top priority. We will work on your balance and core strength. Please remove trip hazards at home (rugs, trailing wires), ensure good lighting, use a grab rail in the bathroom, and wear non-slip footwear. Consider a referral for a home safety assessment."

• Osteoporosis management — GP liaison: "I want to make sure you are on the right medication for your osteoporosis. Long-term steroids significantly reduce bone density. Are you taking a bisphosphonate (e.g., alendronate) or calcium and Vitamin D? If not, I would recommend we flag this to your GP as part of your management."

• Posture and spinal hygiene: "Avoid bending and twisting simultaneously — this is the highest risk movement for your spine. Instead: 'hip hinge' to pick items up (push hips back, keep spine tall). Use a firm chair with good lumbar support. Sleeping on your back or side with a pillow between your knees is safest."

■ Progression Criteria (Before Advancing to Phase 2)

• VAS ≤2/10 during daily activities (from baseline 4/10)

• Able to stand comfortably for ≥15 minutes (from the reported limitation)

• Completing prone extension exercise with 10-second hold × 10 reps pain-free

• TUG score ≤12 seconds (reduced fall risk threshold)

• No deterioration in kyphosis index or neurological screen

■ Long-Term Plan — Weeks 4–12 (Function, Fall Prevention & Independence)

• Progressive strengthening: Advance thoracic extensor programme — increase resistance, reps, and functional positions (standing, integrated with daily activities). Standing resistance band rows and scapular retraction to counteract kyphosis.

• Balance and fall prevention programme: Progress from seated balance exercises → standing on one leg (support available) → tandem stance → stepping exercises. Consider referral to community falls prevention programme (e.g., Otago Exercise Programme — RCT evidence for fall reduction in >65s).

• Functional independence training: Practice sit-to-stand technique, safe bending mechanics (hip hinge), lifting technique, safe stair use. Focus on activities identified in PSFS (e.g., shopping, cooking, dressing).

• Hydrotherapy consideration: Pool-based therapy reduces gravitational load on the spine while allowing progressive strengthening — particularly useful if pain remains a barrier to land-based exercise at this stage.

• Multidisciplinary team involvement: Liaise with GP regarding bone protection medication optimisation; consider occupational therapy for home assessment and assistive devices; dietitian review for calcium/vitamin D intake.